Study of Neutron Star Properties with various hybrid models.

By

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PHYS07202204012

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A thesis submitted to the

Board of Studies in Physical Sciences

In partial fulfillment of requirements

For the Degree of

DOCTOR OF PHILOSOPHY

of

HOMI BHABHA NATIONAL INSTITUTE

July 9, 2026

Homi Bhabha National Institute

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DECLARATION

I, **Debabrata Dey**, hereby declare that the investigations presented in the thesis have been carried out by me. The matter embodied in the thesis is original and has not been submitted earlier as a whole or in part for a degree/diploma at this or any other Institution/University.

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List of Publications for the thesis

Journal

Published

1. *Universality in spacetime ω modes of quarkyonic stars*
   Debabrata Dey, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   Phys. Rev. D 113 (2026) 12, 123034, arXiv:2602.22641v2 [astro-ph].

2. *W-mode oscillation of neutron star in a new relativistic hybrid model*
   Debabrata Dey, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   Nucl. Phys. B 1029 (2026) 117519, arXiv: arXiv:2509.01175v1 [astro-ph].

3. *f-mode oscillations of dark matter admixed quarkyonic neutron star*
   Debabrata Dey, Jeet Amrit Pattanaik, M. Bhuyan, R.N. Panda, S. K. Patra
   JCAP 08 (2025) 003, arXiv:2412.06739v1 [astro-ph].

4. *Dark matter influence on quarkyonic stars: a relativistic mean field analysis*
   Debabrata Dey, H. C. Das, Ankit Kumar, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   JCAP 01 (2025) 056, arXiv:2401.02190v2 [astro-ph].

Work in progress

1. *Core or Halo? Two-Fluid Analysis of Dark Matter-Admixed Quarkyonic Stars in the Multi-Messenger Era*
   Jeet Amrit Pattanaik, **Debabrata Dey**, M. Bhuyan, R.N. Panda, S. K. Patra
   arXiv:2509.06684v1.

2. *Quarkyonic Model for Neutron Star Matter: A Relativistic Mean-Field Approach*
   Ankit Kumar, **Debabrata Dey**, Shamim Haque, Ritam Mallick, S. K. Patra
   arXiv:2304.08223v2.

Conference Proceedings

1. *Dark Matter Effects on f-mode Oscillations in Quarkyonic Stars*
   Debabrata Dey, Jeet Amrit Pattanaik, M. Bhuyan, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 68 (2025) 765-766

2. *Influence of Dark Matter on Neutron Star Curvature within the Quarkyonic Model: A Relativistic Mean Field Approach*
   Debabrata Dey, Jeet Amrit Pattanaik, M. Bhuyan, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 68 (2025) 763-764 .

3. Quarkyonic model for dark matter admixed neutron Star: A RMF perspective
   Debabrata Dey, H. C. Das, Ankit Kumar, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 67 (2024) 807-808.

4. *Influence of Dark Matter on Neutron Star Curvature within the Quarkyonic Model: A Relativistic Mean Field Approach*
   Debabrata Dey, Jeet Amrit Pattanaik, M. Bhuyan, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 68 (2025) 763-764 .

5. Tidal deformability of dark matter admixed neutron star within quarkyonic model in the conjunction of relativistic mean field formalism
   Debabrata Dey, H. C. Das, Ankit Kumar, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 67 (2024) 805-806.

6. f-Mode Universal Relation of Dark Matter Admixed Quarkyonic Stars
   Debabrata Dey, Jeet Amrit Pattanaik, M. Bhuyan, R.N. Panda, S. K. Patra
   Springer Proc.Phys. 432 (2026) 231-234 .

7. Quarkyonic model for dark matter admixed neutron Star: A RMF perspective
   Debabrata Dey, H. C. Das, Ankit Kumar, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 67 (2024) 807-808

8. Tidal deformability of dark matter admixed neutron star within quarkyonic model in the conjunction of relativistic mean field formalism

Jeet Amrit Pattnaik, R. N. Panda, **D. Dey**, Ankit Kumar, S. K. Patra, H. C. Das

DAE Symp.Nucl.Phys. 67 (2024) 805-806.

Talk/Poster

1. **Talk:** *f-Mode Universal Relation of Dark Matter Admixed Quarkyonic Stars*
   26th DAE-BRNS High Energy Physics Symposium 2024,(In Person).

2. **Talk:** *w-mode oscillation of quarkyonic star*
   National Workshop on Recent Advances in Quantum Dynamics, Nuclear and Astrophysics, Institute Of Physics, Bhubaneswar; February 16-17, 2026 (In Person).

3. **Poster:** *Dark Matter Effects on f-mode Oscillations in Quarkyonic Stars*
   68th DAE Symposium on Nuclear Physics, December 7-11, 2024 .

4. **Poster:** *Influence of Dark Matter on Neutron Star Curvature within the Quarkyonic Model: A Relativistic Mean Field Approach*
   68th DAE Symposium on Nuclear Physics, December 7-11, 2024.

5. **Poster:** *Quarkyonic model for dark matter admixed neutron Star: A RMF perspective*
   67th DAE Symposium on Nuclear Physics, December 9-23, 2023.

6. **Poster:** *Tidal deformability of dark matter admixed neutron star within quarkyonic model in the conjunction of relativistic mean field formalism*
   : 67th DAE Symposium on Nuclear Physics, December 9-23, 2023.

(Debabrata Dey)

DEDICATIONS

Dedicated To Maa, Bapi and Bhai

ACKNOWLEDGMENTS

I feel truly grateful to have the opportunity to acknowledge the people who have been a part of my academic journey. First and foremost, I express my sincere respect and gratitude to the soldiers of India and to every taxpayer of the country, whose contributions have made it possible for me to pursue research in a safe and supportive environment.

I am deeply indebted to my Ph.D. supervisor, Prof. Debottam Das, and my co-supervisor, Retd. Prof. Suresh Kumar Patra, for their constant guidance, encouragement, and for giving me the opportunity to pursue research during my doctoral studies. Their mentorship has been invaluable in shaping both my scientific understanding and my approach to research.

I would also like to express my heartfelt gratitude to the members of my Doctoral Committee—Prof. Sanjib Kumar Agarwalla, Prof. Aruna Kumar Nayak, Prof. Manimala Mitra, and Prof. Prafulla Kumar Panda—for their insightful comments and constructive suggestions during my annual review seminars. Their valuable feedback helped me overcome several misconceptions and significantly broadened my understanding of physics.

I am fortunate to have worked with wonderful group members and collaborators. I sincerely thank Harishda for the many stimulating discussions during the early days of my Ph.D. A special word of thanks goes to Jeet, from whom I learned numerous practical research skills and with whom I found not only an excellent collaborator but also a true friend. Our countless conversations over coffee, ranging from physics to life and everything beyond, remain among my most cherished memories.

I would also like to thank the seniors at the Institute of Physics—Arpanda, Abhishekda, Manishda, Biswajitda, Kamleshda, and Kunduda—who never hesitated to extend their support, whether in academic matters or personal life. The late-night rooftop addas at the IOP New Hostel were filled with thoughtful conversations that taught me many lessons and, in many ways, shaped my perspective on life.

My heartfelt thanks also go to all my batchmates, each of whom brought a unique way of thinking about physics and enriched my learning experience. I am especially grateful to Rahul Puri for the many stimulating discussions on physics, to Subhalaxmi Rout for her friendship and support, and to Nevin Noble, whose enthusiasm and vibrant personality made every interaction memorable.

Finally, I owe my deepest gratitude to my family. My Maa and Bapi have been my greatest source of unconditional love, unwavering support, and endless encouragement throughout this uncertain and challenging journey. Their sacrifices and faith in me have been the foundation of everything I have achieved. I am equally grateful to my brother for always standing by my side and being a constant source of strength whenever I needed him.

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CHAPTER 1

Introduction

1.1 Strong Interaction

The strong interaction is one of the four fundamental forces of nature and is responsible for binding quarks into hadrons and, at a larger scale, binding protons and neutrons together to form atomic nuclei. It is described within the Standard Model of particle physics by Quantum Chromodynamics (QCD), a non-Abelian gauge theory built on the color charge carried by quarks and gluons. Unlike the electromagnetic interaction, QCD exhibits two defining features: asymptotic freedom, whereby quarks behave as nearly free particles at short distances (high energies), and confinement, whereby the interaction grows so strong at larger distances that free quarks and gluons are never observed in isolation. This interplay makes the strong interaction uniquely challenging to study, since perturbative techniques that work well at high energies break down in the low-energy regime relevant to nuclear physics.

At the energy scales relevant to nuclear structure, the strong interaction is more conveniently described not by quarks and gluons directly, but by an effective interaction between nucleons (protons and neutrons), historically modeled through meson-exchange theories and, more recently, through chiral effective field theory. This effective nucleon-nucleon interaction retains the essential residual effects of the underlying

QCD dynamics and provides the foundation upon which nuclear structure and, ultimately, the physics of dense stellar matter can be understood. Establishing a reliable and consistent description of the strong interaction across these scales is therefore a prerequisite for the topics discussed in the remainder of this thesis.

1.2 Nucleus

The atomic nucleus is a self-bound, many-body quantum system governed by the residual strong interaction between its constituent nucleons. Despite decades of theoretical and experimental progress, a complete and unified description of nuclear structure remains an open problem, owing to the complexity of the nuclear force, the many-body nature of the system, and the wide range of phenomena nuclei exhibit — from simple shell-like behavior in light and near-magic nuclei to collective excitations, deformation, and clustering in heavier or exotic systems.

Nuclear models such as the shell model, mean-field approaches, and ab initio many-body methods each capture different aspects of nuclear behavior, and their predictions become increasingly important when extrapolated toward regions of the nuclear chart that are difficult or impossible to probe experimentally, such as extremely neutron-rich matter. Understanding how nucleons organize themselves under the strong force, how nuclear matter responds to changes in density and isospin asymmetry, and how the nuclear equation of state (EOS) behaves under extreme conditions are central questions that connect terrestrial nuclear physics to astrophysical phenomena. This connection forms the conceptual bridge to the final subject of this thesis.

1.3 Neutron Star

Neutron stars represent one of the most extreme environments in which the strong interaction and nuclear matter can be studied, offering a natural laboratory that is entirely inaccessible on Earth. Formed in the aftermath of core-collapse supernovae, neutron stars compress roughly a solar mass of matter into a radius of only about ten kilometers, reaching central densities several times that of ordinary atomic nuclei.

Under these conditions, matter is expected to transition through a sequence of exotic phases — from a neutron-rich outer crust, through increasingly dense nuclear matter, to a core whose composition remains uncertain and may include hyperons, meson condensates, or deconfined quark matter.

The properties of neutron stars — their mass-radius relationship, maximum mass, cooling behavior, and tidal deformability during binary mergers — are directly governed by the nuclear equation of state, which is in turn rooted in the strong interaction between nucleons. Recent multi-messenger observations, including precise pulsar mass measurements and gravitational-wave signals from neutron star mergers such as GW170817, have opened an unprecedented observational window into this regime, allowing theoretical predictions derived from nuclear physics to be directly confronted with astrophysical data. This convergence of nuclear theory and astrophysical observation motivates the work presented in this thesis: to use our understanding of the strong interaction and nuclear structure to constrain the properties and internal composition of neutron stars.

1.4 Conventions

- Greek indices run from 0 to 3 and denote space–time components (with 0 being the time component), Latin indices run from 1 to 3 and denote spatial components. We use the Einstein sum convention for these indices.

- Latin indices in roman typeset, e.g. $\mathrm{n}, \mathrm{p}, \mathrm{x}, \ldots$, are constituent indices. We place them liberally upstairs or downstairs in order to avoid cluttering and they do not obey any kind of summation convention.

- The signature of the metric is $(-, +, +, +)$ which means that time-like vectors have a negative length.

- Unless stated otherwise, we carry out our calculations in units in which $G = c = M_\odot = 1$. Whenever appropriate, we quote physical quantities in cgs-units; occasionally we use MeV.

We plan the thesis in the following fashion. In Chapter 2, we introduce the theoretical framework for neutron star

the quantum mechanical approach for constructing the formalism of neutrino oscillations. We provide fundamental insights into two-flavor and three-flavor neutrino oscillations, both in vacuum and Earth's matter. Furthermore, we conduct an in-depth exploration of the degeneracies associated with the neutrino oscillation parameters and highlight the prevailing tensions among various neutrino oscillation experiments. Chapter 3 presents a comprehensive overview of the evolution of long-baseline experiments, tracing their trajectory from inception to their current advancements. In Chapter 4, we illustrate how the contemporary and next-generation long-baseline (LBL) neutrino experiments hold immense potential in probing the deviation of the atmospheric mixing angle (θ_{23}) from maximal mixing, resolving the octant ambiguity ($\theta_{23} > 45^\circ$ or $< 45^\circ$), and significantly enhancing the precision of the oscillation parameters θ_{23} and Δm^2_{31}. We particularly emphasize the role of the upcoming Deep Underground Neutrino Experiment (DUNE) in surpassing present measurements within the three-neutrino framework. Chapter 5 delves into the synergistic interplay between DUNE and T2HK (Tokai-to-Hyper-Kamiokande) in resolving degeneracies among neutrino oscillation parameters. We investigate how the combined sensitivity of these experiments strengthens key aspects of neutrino oscillation phenomenology, including the establishment of non-maximal θ_{23}, exclusion of the incorrect octant, and precision measurements of the 2-3 oscillation parameters of the PMNS matrix. These pursuits are particularly compelling as neutrino oscillation research transitions into the precision era. Additionally, we demonstrate how the complementary features of DUNE and T2HK facilitate an enhanced physics reach, achieving substantial sensitivity at lower exposures compared to their individual baseline configurations. Lastly, in Chapter 6, we explore how the next-generation LBL experiments can uncover signatures of the flavor-dependent long-range neutrino interactions within the neutrino sector. Specifically, we analyze how these experiments can probe the presence of the gauged $(L_e - L_\mu)$, $(L_e - L_\tau)$, and $(L_\mu - L_\tau)$ symmetries through the aforementioned oscillation phenomenology, thereby establishing compelling evidence for such interactions with a high degree of confidence. Finally, Chapter 7 ends this thesis with some insightful concluding remarks, encapsulating the key takeaways, and their importance in the neutrino oscillation frontier.

CHAPTER 2

Theoretical framework

Neutrino oscillation is one of the pioneering discoveries in the history of particle physics. It has opened a new horizon for the community to utilize this second most abundant particle to reveal several mysteries of the Universe [1]. The cornerstone of the neutrino oscillation was first laid in the Super-Kamiokande detector (Japan) in 1998 [2] by providing experimental evidence in their data. Non-zero neutrino mass is the first experimental proof of BSM physics, for which the Nobel Prize was awarded to Takaaki Kajita and Arthur McDonald in 2015 [3, 4].

Before entering into the formalism of neutrino oscillation, it is essential to know why neutrinos show the flavor transition (named "neutrino oscillation") [5, 6], whereas their charged counterparts do not show oscillations. Neutrino oscillation is a mass-induced phenomenon. Due to the non-zero value of neutrino mass and the non-degenerate values of the mass eigenstates [7], the flavor eigenstates and mass eigenstates of neutrinos are quite different. The tiny mass-squared differences between the neutrino mass eigenstates ($\sim 10^{-5}$ to $10^{-3}\,\mathrm{eV}^2$) protect the coherence of the superposition [8, 9] of mass eigenstates to the flavor states. In contrast, the mass differences between the leptons are so huge ($\sim$ 100 MeV to GeV scale) that any superposition of the mass eigenstates of the leptons decoheres almost immediately before their significant propagation.

This chapter is planned in the following way. Section 2.1 outlines the fundamental theoretical framework of neutrino oscillations in the vacuum, examining the formalism in the two-flavor framework. In section 2.5.1, the extension of this formalism in the three-flavor landscape is described. In section ??, we introduce matter-effect in our discussion, incorporating standard neutrino-matter interactions within both two-flavor and three-flavor scenarios. Section ?? provides an overview of several relevant neutrino oscillation experiments and their contributions to measuring oscillation parameters. In section ??, we present the current status of the six oscillation parameters and the so-called eight-fold degeneracies, discussing their experimental determination. Finally, in section ??, we provide a concluding overview of this chapter.

2.1 Neutron star Matter

The E-RMF formalism has demonstrated remarkable robustness in its ability to accurately describe both finite nuclei and infinite nuclear matter. The Lagrangian density is modeled by considering the interactions between different mesons and nucleons, including the self and cross-coupling among them. The parameters of the E-RMF model are determined by fitting with different experimental and empirical data. In literature, more than 200 parameters have been developed to reproduce the different experimental/observational data[? ? ? ? ? ? ? ? ? ? ?]. In this E-RMF model [? ? ? ? ?], the Lagrangian density has the self and cross-coupling between mesons up to fourth order. For completeness, the E-RMF Lagrangian for nucleon-meson-leptons for baryonic matter (BM) system is given as [? ? ?]:

$$\begin{aligned}
\mathcal{L}_{\mathcal{BM}} &= \sum_{i=n,p} \bar{\psi}_i \Bigg\{ \gamma_\nu (i\partial^\nu - g_\omega \omega^\nu - \frac{1}{2} g_\rho \vec{\tau}_i \cdot \vec{\rho}^{\,\nu}) - (M_{nucl.} - g_\sigma \sigma \\
&\quad - g_\delta \vec{\tau}_i \cdot \vec{\delta}) \Bigg\} \psi_i - \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{2} \partial^\nu \sigma \, \partial_\nu \sigma + \frac{1}{2} m_\omega^2 \omega^\nu \omega_\nu \\
&\quad - \frac{1}{4} F^{\alpha\beta} F_{\alpha\beta} + \frac{1}{2} m_\rho^2 \rho^\nu \cdot \rho_\nu - \frac{1}{4} \vec{R}^{\alpha\beta} \cdot \vec{R}_{\alpha\beta} - \frac{1}{2} m_\delta^2 \vec{\delta}^{\,2} \\
&\quad + \frac{1}{2} \partial^\nu \vec{\delta} \, \partial_\nu \vec{\delta} - g_\sigma \frac{m_\sigma^2}{M} \left(\frac{\kappa_3}{3!} + \frac{\kappa_4}{4!} \frac{g_\sigma}{M} \sigma \right) \sigma^3
\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{2}\frac{g_\sigma\sigma}{M}\left(\eta_1+\frac{\eta_2}{2}\frac{g_\sigma\sigma}{M}\right)m_\omega^2\omega^\nu\omega_\nu+\frac{\zeta_0}{4!}g_\omega^2(\omega^\nu\omega_\nu)^2\\
&+\frac{1}{2}\eta_\rho\frac{m_\rho^2}{M}g_\sigma\sigma(\vec{\rho}^{\,\nu}\cdot\vec{\rho}_\nu)-\Lambda_\omega g_\omega^2 g_\rho^2(\omega^\nu\omega_\nu)(\vec{\rho}^{\,\nu}\cdot\vec{\rho}_\nu)\\
&+\sum_{j=e^-,\mu}\bar{\phi}_j(i\gamma_\nu\partial^\nu-m_j)\phi_j. \qquad (2.1)
\end{aligned}
$$

The nucleon wave functions, denoted by ψ_i, represent both neutrons and protons, while the last term in the expression accounts for the non-interacting leptonic part, namely electrons and muons. The nucleon mass is denoted by $M_{nucl.}$ (approximately 939 MeV), and specific masses and coupling constants are assigned to various mesons, including the sigma (m_σ, g_σ, κ_3, κ_4), omega (m_ω, g_ω, ζ_0, η_1, η_2), rho (m_ρ, g_ρ, η_ρ, Λ_ω), and delta (m_δ, g_δ) mesons. The field strength tensors, $F^{\alpha\beta}$ and $\vec{R}^{\alpha\beta}$, are employed for the omega and rho mesons, respectively. To facilitate further calculations, we adopt the relativistic mean-field (RMF) approximation, where meson fields are replaced with their average values. This simplifies computations, especially in cases of uniform static matter, where spatial and temporal derivatives of mesons can be disregarded. Maintaining translational and rotational invariance, as well as isotropy of nuclear matter, it is crucial for ensuring the accuracy and consistency of calculations. In this approximation, only the time-like components of the isovector field and the isospin 3 component of the mesonic field are considered significant. [? ? ? ? ? ? ?]. The values of the parameter sets used in the present work are given in Table ??, and their nuclear matter properties at saturation density are listed in Table ??. The energy density ($\mathcal{E}_{\rm BM}$) and pressure ($P_{\rm BM}$) for BM system can be computed from the Lagrangian (Eq. 2.1) using the stress-energy tensor, yielding as [?]:

$$
\begin{aligned}
\mathcal{E}_{\rm BM} \;=\; &\sum_{i=n,p}\frac{g_s}{(2\pi)^3}\int_0^{k_{f_i}} d^3k\,\sqrt{k^2+M_i^{*2}}\\
&+n g_\omega\,\omega+m_\sigma^2\sigma^2\left(\frac{1}{2}+\frac{\kappa_3}{3!}\frac{g_\sigma\sigma}{M_{\rm nucl.}}+\frac{\kappa_4}{4!}\frac{g_\sigma^2\sigma^2}{M_{\rm nucl.}^2}\right)\\
&-\frac{1}{4!}\zeta_0\,g_\omega^2\,\omega^4-\frac{1}{2}m_\omega^2\,\omega^2\left(1+\eta_1\frac{g_\sigma\sigma}{M_{\rm nucl.}}+\frac{\eta_2}{2}\frac{g_\sigma^2\sigma^2}{M_{\rm nucl.}^2}\right)\\
&+\frac{1}{2}n_3\,g_\rho\,\rho-\frac{1}{2}\left(1+\frac{\eta_\rho g_\sigma\sigma}{M_{\rm nucl.}}\right)m_\rho^2
\end{aligned}
$$

$$-\Lambda_\omega\, g_\rho^2\, g_\omega^2\, \rho^2\, \omega^2 + \frac{1}{2} m_\delta^2\, \delta^2$$
$$+ \sum_{j=e^-,\mu} \frac{g_s}{(2\pi)^3} \int_0^{k_{f_j}} \sqrt{k^2 + m_j^2}\, d^3k, \tag{2.2}$$

and

$$\begin{aligned}
P_{\rm BM} &= \sum_{i=n,p} \frac{g_s}{3(2\pi)^3} \int_0^{k_{f_i}} d^3k\, \frac{k^2}{\sqrt{k^2 + M_i^{*2}}} \\
&\quad - m_\sigma^2 \sigma^2 \left(\frac{1}{2} + \frac{\kappa_3}{3!} \frac{g_\sigma \sigma}{M_{\rm nucl.}} + \frac{\kappa_4}{4!} \frac{g_\sigma^2 \sigma^2}{M_{\rm nucl.}^2} \right) + \frac{1}{4!} \zeta_0\, g_\omega^2\, \omega^4 \\
&\quad + \frac{1}{2} m_\omega^2 \omega^2 \left(1 + \eta_1 \frac{g_\sigma \sigma}{M_{\rm nucl.}} + \frac{\eta_2}{2} \frac{g_\sigma^2 \sigma^2}{M_{\rm nucl.}^2} \right) \\
&\quad + \frac{1}{2} \left(1 + \frac{\eta_\rho g_\sigma \sigma}{M_{\rm nucl.}} \right) m_\rho^2\, \rho^2 - \frac{1}{2} m_\delta^2\, \delta^2 + \Lambda_\omega g_\rho^2 g_\omega^2 \rho^2 \omega^2 \\
&\quad + \sum_{j=e^-,\mu} \frac{g_s}{3(2\pi)^3} \int_0^{k_{f_j}} \frac{k^2}{\sqrt{k^2 + m_j^2}}\, d^3k,
\end{aligned} \tag{2.3}$$

where $g_s = 2$ is the spin degeneracy of nucleons. The effective mass of nucleon (i= n, p) is defined as:

$$M_i^\star = M_{\rm nucl.} + g_\sigma \sigma_0 \pm g_\delta \delta_0. \tag{2.4}$$

Here, k_{fi} and k_{fj} are the Fermi momenta of the nucleons and leptons, respectively. The baryonic and the isovector densities are denoted as n and n_3. The σ_0 and δ_0 are the mean meson fields of sigma and delta mesons.

2.2 Quarkyonic Star Matter

Now, we discuss the quarkyonic model as proposed by Mclerran and Reddy [?]. In this phenomenological model, it is suggested that at the core of the NS, where the density is several times higher than the nuclear saturation density, the nucleons break into quarks. The defining characteristic of their model is its capacity to influence the correlation between mass and radius, specifically predicting a higher mass due to the stiffening of EOS as compared to baryonic. The onset of the quarkyonic phase from baryonic is marked by a swift rise in pressure, which is due to the occupation of quarks in lower momentum states as the baryon density reaches a critical value, termed transition density. This results in treating the low momentum degrees of freedom as quarks while high momenta degrees of freedom near the Fermi surface as nucleons. The momentum states near the Fermi surfaces are in order of QCD confinement scale Λ_{cs} and hence lead to the formation of nucleons, which are the bound states of quarks.

The schematic model of Mclerran and Reddy [?] is further developed and modified by Jhao and Lattimer [?]. They incorporated the beta-equilibrium condition for quarkyonic matter along with charge neutrality for neutron star matter (NSM). The nucleons interact via density-dependent potential, which is fitted to select properties of uniform NM. In addition, they established chemical equilibrium among nucleons and quarks in order to establish the relation between nucleon momenta $k_{f_{(n,p)}}$ and quark $k_{f_{(u,d)}}$, which marks the distinctiveness of the modified quarkyonic model. Since nucleons occupy a Fermi shell in the quarkyonic matter model, it has a minimum momentum $k_{0_{(n,p)}}$ and Fermi momentum $k_{f_{(n,p)}}$. The d and u quark Fermi momentum are k_{f_d} andk_{f_u} respectively. The conservation law of baryon density implies the relation in [?];

$$\begin{aligned} n &= n_n + n_p + \frac{n_u + n_d}{3} \\ &= \frac{g_s}{6\pi^2}\left[(k_{f_n}^3 - k_{0_n}^3) + (k_{f_p}^3 - k_{0_p}^3) + \frac{(k_{f_u}^3 + k_{f_d}^3)}{3}\right]. \end{aligned} \tag{2.5}$$

Here, n_n, n_p, n_u, n_d are the neutron, proton, up and down quark baryon densities. Also, the charge neutrality condition among nucleons, quarks, and leptons leads to;

$$n_p + \frac{2n_u}{3} - \frac{n_d}{3} = n_{e^-} + n_\mu. \tag{2.6}$$

Here, n_{e^-} and n_μ are electron and muon densities respectively.

The $k_{f0(n,p)}$ is related to the transition momentum of nucleons $k_{t_{(n,p)}}$ corresponding to transition density from nucleonic to quarkyonic matter at baryon density $n = n_t$ by the following expression [?]

$$k_{0(n,p)} = (k_{f_{(n,p)}} - k_{t_{(n,p)}})\left[1 + \frac{\Lambda_{\rm cs}^2}{k_{f_{(n,p)}} k_{t_{(n,p)}}}\right]. \tag{2.7}$$

$$\begin{aligned} k_{0(n,p)} &= 0 \\ &= (k_{f_{(n,p)}} - k_{t_{(n,p)}})\left[1 + \frac{\Lambda_{\rm cs}^2}{k_{f_{(n,p)}} k_{t_{(n,p)}}}\right]. \end{aligned} \tag{2.8}$$

. (2.9)

At n_t which is in the order of approximately $2.0 - 3.0$ times the saturation density, the nucleons will overlap with each other, and quarks will drip out from the nucleon to form a fermi shell of high momenta nucleon and low momenta quarks. Here, the $\Lambda_{\rm cs}$ is the QCD confinement scale. This acts as the cut-off momentum scale between nucleons and quarks. The low momentum degrees of freedom i.e. the momentum less than $\Lambda_{\rm cs}$ are treated as quarks while higher momenta are treated as nucleons. The value of k_0 cannot be negative because the model construction ensures that the expression for k_0 applies only in the regime where $k_{f(n,p)} > k_{t(n,p)}$. In this regime, which is above transition density the terms ensure a non-negative value for $k_{0(n,p)}$. Below the transition density, the expression is not applicable, and $k_{0(n,p)}$ is set to zero.

Here, Strong interaction equilibrium ensures that at each fixed baryon density, the Fermi gas has its lowest possible energy. It is equivalent to the condition of the chemical equilibrium among nucleons and quarks given by the relations [**?**];

$$\mu_n = \mu_u + 2\mu_d, \tag{2.10}$$

$$\mu_p = 2\mu_u + \mu_d, \tag{2.11}$$

where μ_n, μ_p, μ_u, and μ_d are the chemical potentials of neutron, proton, up quark, and down quark respectively. The relationship among the chemical potentials and their respective Fermi momenta is given by the following expression:

$$\begin{aligned}\mu_{n,p} = \frac{\partial \varepsilon_B}{\partial n_{n,p}} = (1 - K_{n,p})^{-1} \times \\ \times \left(\sqrt{m_{n,p} + k^2_{f(n,p)}} - K_{n,p}\sqrt{m^2_{n,p} + k^2_{0(n,p)}}\right) \\ + \frac{\partial[(n_n + n_p)V(n_n, n_p)]}{\partial n_{n,p}},\end{aligned} \tag{2.12}$$

where

$$\begin{aligned}K_{n,p} &= \left(\frac{k_{0(n,p)}}{k_{f(n,p)}}\right)^2 \frac{dk_{0(n,p)}}{dk_{f(n,p)}} \\ &= \left(\frac{k_{0(n,p)}}{k_{f(n,p)}}\right)^2 \left[1 + \left(\frac{\Lambda_{\rm cs}}{k_{f(n,p)}}\right)^2\right].\end{aligned} \tag{2.13}$$

and

$$\mu_{u,d} = \sqrt{m_{u,d} + k^2_{f(u,d)}}. \tag{2.14}$$

Here $V(n_n, n_p)$ is the mesonic mean field potential energy among nucleons. In addition to strong interaction equilibrium, the energy of the Fermi gas is further minimized due to the beta-equilibrium condition under the constraint of charge neutrality. This results in establishing chemical equilibrium between neutron, proton, electron, and muon, as indicated[**?** **?**];

$$\begin{aligned}\mu_n &= \mu_p + \mu_{e^-}, \\ \mu_\mu &= \mu_{e^-}.\end{aligned} \tag{2.15}$$

Here, μ_μ, μ_{e^-} are chemical potential of muon and electron respectively. These leptons chemical potential are related with their respective Fermi momenta by similar equation as in 2.14.

Also, one crucial aspect of the model is the up and down quark masses, which are not independent variables but dependent on the beta equilibrium condition of NS matter at transition density n_t via;

$$m_u = \frac{2}{3}\mu_{t_p} - \frac{1}{3}\mu_{t_n}, \qquad m_d = \frac{2}{3}\mu_{t_n} - \frac{1}{3}\mu_{t_p}, \tag{2.16}$$

where μ_{t_n} and μ_{t_p} are the chemical potential of neutron and proton at n_t. In the context of our model, the quark number density is given by :

$$n_q = \frac{g_s}{6\pi^2}\frac{N_c}{3}k_{fq}^3,$$

where $q = (u, d)$, $g_s = 2$ is the spin degeneracy factor, $N_c = 3$ is the color degeneracy factor, and k_{fq} is the quark Fermi momentum. To ensure zero quark density at the quark-hadron transition, the proper relationship between the quark Fermi momenta and chemical potentials must be established. The quark number density n_q becomes zero when the quark Fermi momentum k_{fq} is zero. This occurs at the transition point where the chemical potential conditions for the hadrons and quarks align. At the quark-hadron transition, the chemical potentials of the quarks and hadrons must satisfy the conditions for beta equilibrium and charge neutrality. These conditions ensure that the chemical potentials are balanced, leading to the proper phase transition. The chemical potentials of quarks in beta equilibrium are given by

$$\mu_q = \sqrt{m_q^2 + k_{fq}^2},$$

where m_q is the mass of the quarks and μ_q is the chemical potential of the quarks. To achieve zero quark density at the transition, the Fermi momenta of the quarks must be such that $k_{fq} = 0$ at $\mu_q = \mu_{q,\text{transition}}$. Below the transition density the beta equllibrium condition of npeμ matter is given by

$$\mu_n = \mu_p + \mu_e,$$

$$\mu_\mu = \mu_e,$$

This approach ensures that the up and down quark masses are defined self-consistently, taking into account the beta equilibrium among nucleons and quarks, which is essential for accurately describing the physical state of matter at the high densities characteristic of NSs. Consequently, the up and down quark masses, typically found to be $m_u \approx 241$ MeV and $m_d \approx 391$ MeV, differ significantly from the current quark masses due to the effects of the dense medium in the NS interior [**?**]. Since the quarks are considered as non-interacting fermion gas, their energy density $\mathcal{E}_{\rm QM}$ and pressure $P_{\rm QM}$ can be written as [**?**];

$$\mathcal{E}_{\rm QM} = \sum_{q=u,d} \frac{g_s N_c}{(2\pi)^3} \int_0^{k_{fq}} k^2 \sqrt{k^2 + m_q^2}\, d^3k, \tag{2.17}$$

$$P_{\rm QM} = \mu_u n_u + \mu_d n_d - \mathcal{E}_{\rm QM}\,. \tag{2.18}$$

where $N_c = 3$ is the color degeneracy of quarks, m_q is the mass of the quark (j = u, d) and k_{fq} is the Fermi momenta of quarks.

2.3 Dark Matter

In this, we choose a simple DM model, where DM particles interact with nucleons and quarks by exchanging Higgs. The Lagrangian density for this interaction is given by [**? ? ? ?**];

$$\begin{aligned} \mathcal{L}_{\rm DM} &= \bar{\chi}\left[i\gamma^\mu \partial_\mu - M_\chi + yh\right]\chi + \frac{1}{2}\partial_\mu h \partial^\mu h \\ &\quad -\frac{1}{2}M_h^2 h^2 + f\frac{M_{\rm nucl.}}{v}\bar{\psi} h \psi, \end{aligned} \tag{2.19}$$

where χ and ψ are the wave functions for the DM and nucleons, respectively and h is the Higgs field. The interaction between Higgs and nucleon is Yukawa type, having its coupling constant f. The f is the proton-Higgs form factor. Here, we consider that the Neutralino is a DM particle having mass M_χ= 200 GeV. The magnitude for the y and f are taken as 0.07 and 0.35, respectively, and it has been constrained using different experimental/empirical data [**?**]. The Higgs mass (M_h) and its vacuum value (v) are 125 and 246 GeV, respectively. In the Standard Model, the Higgs field couples directly to fundamental particles such as quarks. However, in our discussion, when we refer to the Higgs-nucleon coupling, we are considering an effective interaction that arises at the low-energy scale relevant for nucleons. Nucleons are composite particles made up of quarks bound together by gluons. The effective Higgs-nucleon coupling comes from the sum of the Higgs interactions with the quarks inside the nucleon. The nucleon mass itself is largely derived from the quark masses and the strong interaction dynamics, both of which are influenced by the Higgs mechanism. Thus, the coupling we refer to is not a direct fundamental interaction between the Higgs field and the nucleon, but rather an effective coupling that encapsulates the interactions of the Higgs field with the quark constituents of the nucleon. This effective coupling is crucial for calculating the cross sections relevant for DM direct detection experiments, where DM particles interact with nucleons rather than free quarks [?]

The Euler Lagrange equation of motion for DM particle (χ) and Higgs boson (h) can be derived from Lagrangian as [?],

$$\begin{aligned}\Big(i\gamma^\mu\partial_\mu - M_\chi + yh\Big)\chi &= 0,\\ \partial_\mu\partial^\mu h + M_h^2 h &= y\bar{\chi}\chi + \frac{fM_{\text{nucl.}}}{v}\bar{\psi}\psi,\end{aligned} \tag{2.20}$$

respectively. Applying mean-field approximation, we get,

$$h_0 = \frac{y\langle\bar{\chi}\chi\rangle + f\frac{M_{\text{nucl.}}}{v}\langle\bar{\psi}\psi\rangle}{M_h^2},$$

where h_0 is the mean Higgs field. We define $M_\chi^\star$ is the DM effective mass given as,

$$M_\chi^\star = M_\chi - yh_0. \tag{2.21}$$

The DM scalar density (ρ_s^{DM}) is

$$\rho_s^{\text{DM}} = \langle\bar{\chi}\chi\rangle = \frac{\gamma}{2\pi^2}\int_0^{k_f^{\text{DM}}} dk\,\frac{M_\chi^\star}{\sqrt{M_\chi^{\star 2} + k^2}}, \tag{2.22}$$

where k_f^{DM} is the Fermi momentum for DM and $\langle\bar{\psi}\psi\rangle$ is the scalar density of the baryonic sector. $\gamma = 2$ is the spin degeneracy factor for fermionic DM. The energy density and pressure for the DM can be calculated with the mean-field approximation, which is given by [? ? ? ?];

$$\mathcal{E}_{\rm DM} = \frac{\gamma}{(2\pi)^3}\int_0^{k_f^{\rm DM}} d^3k\sqrt{k^2+(M_\chi^\star)^2}+\frac{1}{2}M_h^2h_0^2, \tag{2.23}$$

$$P_{\rm DM} = \frac{\gamma}{3(2\pi)^3}\int_0^{k_f^{\rm DM}} \frac{d^3k k^2}{\sqrt{k^2+(M_\chi^\star)^2}}-\frac{1}{2}M_h^2h_0^2, \tag{2.24}$$

where $k_f^{\rm DM}$ is the DM Fermi momentum and $M_\chi^\star$ represent DM effective mass defined in (2.21).

We calculate the spin-independent scattering cross-section of the nucleons with DM using the relation given as [?]

$$\sigma_{\rm SI} = \frac{y^2 f^2 M_{\rm nucl.}^2}{4\pi}\frac{\mu_r}{v^2 M_h^2}, \tag{2.25}$$

where μ_r is the reduced mass. The calculated cross-section is found to be 9.70×10^{-46} cm^2, which is almost consistent with XENON-1T [?], PandaX-II [?], PandaX-4T [?] and LUX [?] for the DM mass order of hundred GeV. The LHC had also given a limit on the WIMP-nucleon scattering cross-section in the range from $10^{-40}-10^{-50}$ cm^2 [?]. Thus our model also satisfies the LHC limit. Therefore in the present calculations, we constrained the value of y from both the direct detection experiments and the LHC results. Assuming the nuclear saturation density of nucleons (n) is 10^3 times larger than the average DM density ($n_{\rm DM}$), which implies the ratio of the DM to NS mass as $M_{DM}/M_{NS}\sim\frac{1}{6}$ [?], where M_{NS} is the mass of NS and M_{DM} is the total DM mass inside the NS. Using the nuclear saturation density as $n_0\sim0.16$ fm^{-3}, the DM number density becomes $n_{\rm DM}\sim10^{-3}n_0\sim0.16\times10^{-3}$ fm^{-3}. The $k_f^{\rm DM}$ is obtained from the equation $k_f^{\rm DM}=(3\pi^2 n_{\rm DM})^{1/3}$. Hence the value of $k_f^{\rm DM}$ is ~0.033 GeV.

Generally, DM models include Weakly Interacting Massive Particles (WIMPs) and Feebly Interacting Massive Particles (FIMPs). In both cases, dark matter interacts very weakly with baryonic matter. However, despite these minimal interactions, the presence of dark matter can have a considerable impact on baryonic matter, especially in systems of extremely dense matter, like neutron stars. According to numerous studies [? ? ? ?], the number density of dark matter inside a neutron star is typically around 1000 times smaller than the normal nuclear saturation density density. Thus, the assumption that the nuclear saturation density of nucleons (n) is 1000 times larger than the average dark matter density ($n_{\rm DM}$) is motivated by the idea that dark matter in a neutron star is expected to be much more dilute compared to nucleons distribution, which are packed at very high densities due to degeneracy pressure and gravitational compression in the neutron star interior. This large disparity between nucleon and dark matter densities reflects the expectation that neutron stars are predominantly composed of baryonic matter, while any dark matter contribution is comparatively minimal. To elaborate, we express the mass ratio as,

$$\frac{\rho_{DM}}{\rho_{Total}} = \frac{\frac{M_{DM}}{(4/3)\pi R^3}}{\frac{M_{Total}}{(4/3)\pi R^3}} = \frac{M_{DM}}{M_{Total}} = \frac{n_{\rm DM} m_{\rm DM}}{n m_n + n_{\rm DM} m_{\rm DM}} = \frac{1}{6} = 0.17 = 17\%, \tag{2.26}$$

The ρ_{Total} refers as the combine density of baryons and presence of dark matter inside the neutron star., i.e. $\rho_{Total} = (\rho_{Baryonic} + \rho_{DM})$. Similarly M_{Total} is the total mass of the baryonic matter as well as mass of the dark matter. Taking nucleon density $n \approx 10^3 n_{\rm DM}$, mass of DM $\mathrm{m}_{DM} = 200$ GeV, mass of nucleon $\mathrm{m}_n = 939$ MeV, we get $\frac{M_{DM}}{M_{NS}} = 1/6$. The modified effective masses of nucleon ($M_i^\star, i = n, p$) due to their interaction with the Higgs field :

$$M_i^\star = M_{\rm nucl.} + g_\sigma \sigma_0 \pm g_\delta \delta_0 - \frac{f M_{\rm nucl.}}{v} h_0, \tag{2.27}$$

. Now the total energy and pressure of DM admixed quarkyonic star is given as;

$$\mathcal{E} = \mathcal{E}_{\rm BM} + \mathcal{E}_{\rm QM} + \mathcal{E}_{\rm DM}, \tag{2.28}$$

$$P = P_{\rm BM} + P_{\rm QM} + P_{\rm DM}. \tag{2.29}$$

where $\mathcal{E}_{\mathrm{BM}}$, $\mathcal{E}_{\mathrm{QM}}$, $\mathcal{E}_{\mathrm{DM}}$ are the energy density for the BM, quarkyonic matter and DM and P_{BM}, P_{QM}, P_{DM} are the corresponding pressure respectively.

2.4 Equllibrium state

The equilibrium configuration of a non-rotating neutron star is determined by solving the Einstein field equations for a self-gravitating, spherically symmetric perfect fluid. The spacetime geometry is described by the standard static, spherically symmetric line element:

$$ds^2 = -e^{\nu(r)}\, dt^2 + e^{\lambda(r)}\, dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\phi^2), \tag{2.30}$$

where $\nu(r)$ and $\lambda(r)$ are metric potentials depending only on the radial coordinate r. For a perfect fluid, the energy-momentum tensor is $T^{ab} = (\epsilon + P)u^a u^b + P g^{ab}$, where $\epsilon(r)$ is the energy density, $P(r)$ is the pressure, and u^a is the fluid four-velocity.

$$G_{ab} = 8\pi G T_{ab}, \tag{2.31}$$

$$T^{ab} = (\epsilon + P)u^a u^b + P g^{ab}. \tag{2.32}$$

2.5 Gravitational Mass and Radius

The Einstein field equations, $G_{ab} = 8\pi G T_{ab}$, with this metric and energy-momentum tensor yield the TOV equations [**? ? ?**]:

$$\frac{dm}{dr} = 4\pi r^2 \epsilon(r), \tag{2.33}$$
$$\frac{dP}{dr} = -\frac{G}{r^2}\left[\epsilon(r) + P(r)\right]\left[m(r) + 4\pi r^3 P(r)\right] e^{\lambda(r)}, \tag{2.34}$$

where the mass function $m(r)$ represents the mass enclosed within radius r, and the metric function is given by:

$$e^{-\lambda(r)} = 1 - \frac{2m(r)}{r}, \tag{2.35}$$

$$\frac{d\nu}{dr} = 2\,\frac{(M + 4\pi r^3 p)}{r(r-2M)}. \tag{2.36}$$

Equation (2.33) defines the gravitational mass, while Equation (2.34) governs hydrostatic equilibrium. The term $[m(r) + 4\pi r^3 P(r)]$ reveals pressure's role as a gravitational source in general relativity, a fundamental departure from Newtonian theory. The factor $e^{\lambda(r)}$ accounts for spacetime curvature effects, which become significant in the high-density core. The system is closed by an equation of state $P(\epsilon)$. Integration proceeds from the center ($r = 0$) with boundary conditions $m(0) = 0$ and a central pressure $P(0) = P_c$, outward to the stellar surface R defined by $P(R) = 0$. The total gravitational mass is then $M = m(R)$.

2.5.1 Tidal Deformability

Tidal deformability λ_{tidal} is a key parameter that provides valuable insights into the internal structure of compact celestial objects. It enables us to understand how these objects respond to tidal forces created by a companion star. It is defined mathematically as $\lambda_{\text{tidal}} = -\frac{Q_{ij}}{\mathcal{E}_{ij}}$ where Q_{ij} and $\mathcal{E}_{ij}$ is the induced quadrupole moment Q_{ij} tensor caused by the external tidal field $\mathcal{E}_{ij}$ tensor. For $l = 2$ it is related to dimensionless tidal love number k_2 and radius R by the relation $\lambda_{\text{tidal}} = \frac{2}{3}k_2R^5$ [**?** **?**]. It's worth noting that the tidal Love number, k_2, is intimately connected to the internal structure and composition of the star and can be measured in gravitational waves emanating from inspiralling binary NSs [**?** **?**]. A dimensionless form of tidal deformability defined as $\Lambda = \frac{\lambda_{\text{tidal}}}{M^5}$ is introduced due to its convenience in comparing the deformability of NSs of different masses.

2.6 Moment Of Inertia

The total moment of inertia in the slow rotating approximation is given by the integral [**?**]

$$I = \frac{8\pi}{3} \int_0^R (\epsilon + p)\, \lambda(r) r^4 \frac{\bar{\omega}(r)}{\Omega}\, dr \tag{2.37}$$

where $\lambda(r) = e^{\lambda}(r) = (1 - 2m(r)/r)^{-1}$ and $\bar{\omega}(r)$ is the rotational drag whose equation is given by:

$$\frac{d}{dr}\left(r^4 j(r) \frac{d\bar{\omega}}{dr}\right) = -4r^3 \frac{dj}{dr}\bar{\omega} \tag{2.38}$$

. Here $j(r) = e^{\nu}(r) + \lambda(r) r^4$, which has the boundary value j(R) = 1 . In the limit of slow rotation, such that the rotation angular velocity $\Omega \leq M/R^2$, $\bar{\omega}$ has the boundary condition $\bar{\omega}(R)/\Omega = 1 - 2I/R^3$. These complex equations are integrated from r = 0 to R, where the pressure drops to a negligible level. This process uses a specific EOS $p = p(\epsilon)$ and a central density $\epsilon(0)$to achieve the NS mass. Consequently, this also determines the radius. By ensuring the boundary conditions are met, the moment of inertia can then be calculated.

2.7 Perturbed state

The neutron star undergoes coupled perturbations in its interior matter distribution and exterior spacetime, where fluctuations of the relativistic fluid are dynamically linked to spacetime curvature through Einstein's field equations. These perturbations encode information about the dense-matter equation of state and the star's internal composition, while the exterior response governs the propagation of gravitational radiation. We briefly outlined the interior and exterior regions of the NS below.

2.7.1 Interior region of the Neutron star

The interior of a nonrotating neutron star is characterized by two fundamental degrees of freedom: the matter perturbations, describing oscillations of the dense nuclear fluid, and the spacetime perturbations, which represent the associated ripples in the gravitational field [**? ?**]. This coupling is essential for modeling stellar oscillations that can emit gravitational waves. The even-parity perturbations in the Regge–Wheeler gauge are described by the metric [?].

$$\begin{aligned} ds^2 = &- e^{\nu}\left(1 + r^{\ell} H_0 Y_{\ell m} e^{i\omega t}\right) dt^2 \\ &+ e^{\lambda}\left(1 - r^{\ell} H_0 Y_{\ell m} e^{i\omega t}\right) dr^2 \\ &+ r^2\left(1 - r^{\ell} K Y_{\ell m} e^{i\omega t}\right) d\Omega^2 \\ &- 2i\omega r^{\ell+1} H_1 Y_{\ell m} e^{i\omega t}\, dt\, dr, \end{aligned} \tag{2.39}$$

where H_0, H_1, and K are radial functions representing the metric perturbations, and ω is the complex oscillation frequency whose real part gives the mode frequency and imaginary part the damping rate due to gravitational wave emission. The fluid perturbations are described by the Lagrangian displacement vector, which characterizes how fluid elements move during oscillations [? ?]:

$$\begin{aligned} \xi^r &= r^{\ell-1} e^{-\lambda/2} W Y_{\ell m} e^{i\omega t}, \\ \xi^{\theta} &= -r^{\ell-2} V \partial_{\theta} Y_{\ell m} e^{i\omega t}, \\ \xi^{\phi} &= -\frac{r^{\ell-2}}{\sin^2\theta} V \partial_{\phi} Y_{\ell m} e^{i\omega t}, \end{aligned} \tag{2.40}$$

where W and V represent the radial and the horizontal (tangential) displacement, respectively. The $Y_{\ell m}(\theta, \phi) = Y_{\ell m}(e^{i\omega t})$ represents the spherical harmonics.

2.8 f-mode eigenvalue problem

The perturbation amplitude W and V are defined by the above equations. The oscillation equation within relativistic cowling approximation is derived by setting $H_0 = H_1 = K = 0$. These assumptions result in:

$$\begin{aligned} \frac{dW}{d\ln r} &= -(l+1)\left[W - l e^{\nu+\lambda/2} U\right] \\ &\quad - \frac{e^{\lambda/2}(\omega r)^2}{c_{ad}^2}\left[U - \frac{d\Phi}{d\ln r}\frac{e^{-\lambda/2}}{(\omega r)^2} W\right], \\ \frac{dU}{d\ln r} &= e^{\lambda/2-\nu}\left[W - l e^{\nu-\lambda/2} U\right] \end{aligned} \tag{2.41}$$

$$+(\frac{1}{c_{\rm eq}^2} - \frac{1}{c_{\rm ad}^2})\frac{d\Phi}{d\ln r}\left[U - \frac{d\Phi}{d\ln r}\frac{e^{-\lambda/2}}{(\omega r)^2}W\right] . \tag{2.42}$$

Here, $W = e^{\lambda/2}r^{1-l}\xi^r$, $U = -e^{-\nu}V$ and $\Phi = 2\nu$. The boundary conditions can be written explicitly as,

$$\left.\frac{W}{U}\right|_{r=0} = le^{\nu|_{r=0}} \tag{2.43}$$

$$\left.\frac{W}{U}\right|_{p=0} = \frac{\omega^2 R^3}{M}\sqrt{1 - \frac{2GM}{R}} . \tag{2.44}$$

Solving these equations gives us the f-mode oscillation frequency of the Neutron star within the relativistic cowling approximation.

2.9 w-mode eigenvalue problem

To formulate a well-posed eigenvalue problem for ω, one must avoid singularities that arise in the perturbation equations. This is achieved by introducing a new variable X, which is related to the Lagrangian pressure variation:

$$\Delta p = -r^\ell e^{-\nu/2} X Y_{\ell m} e^{i\omega t}. \tag{2.45}$$

The system is then reduced to four first-order differential equations for the variables H_1, K, W, and X [**? ?**], which together describe the coupled fluid-spacetime system.

The evolution of these variables is governed by[?]:

$$
\begin{aligned}
r\frac{dH_1}{dr} &= -\left[\ell + 1 + 2be^{\lambda} + 4\pi r^2 e^{\lambda}(p-\epsilon)\right] H_1 \\
&\quad + e^{\lambda}\left[H_0 + K - 16\pi(\epsilon+p)V\right], \\
r\frac{dK}{dr} &= H_0 + (\ell+1)H_1 + \left[e^{\lambda}Q - \ell - 1\right] K \\
&\quad - 8\pi(\epsilon+p)e^{\nu/2}W, \\
r\frac{dW}{dr} &= -(\ell+1)\left[W + \ell e^{\nu/2}V\right] \\
&\quad + r^2 e^{\lambda/2}\left[\frac{X}{(\epsilon+p)c_s^2}e^{-\nu/2} + \frac{H_0}{2} + K\right], \\
r\frac{dX}{dr} &= -\ell X + \frac{\epsilon+p}{2}e^{\nu/2}\Bigg\{(3e^{\lambda}Q - 1)K \\
&\quad - \frac{4(\ell^2+\ell-2)e^{\lambda}Q}{r^2}V + (1-e^{\lambda}Q)H_0 \\
&\quad + \left(r^2\omega^2 e^{-\nu} + \ell^2 + \ell - 2\right) H_1 \\
&\quad + \Bigg[2\omega^2 e^{\lambda/2-\nu} - 8\pi(\epsilon+p)e^{\nu/2} \\
&\qquad + r^2\frac{d}{dr}\left(\frac{e^{-\nu/2}g}{r^2}\right)\Bigg]W\Bigg\},
\end{aligned}
\tag{2.46}
$$

where $Q = M + 4\pi r^3 p$ represents the gravitational mass including pressure contributions, $b = GM/r$ is the compactness, and c_s is the adiabatic sound speed. The first equation governs the evolution of the metric perturbation H_1, which is coupled to fluid variables through V and W. The second equation describes the curvature perturbation K, which sources and is sourced by both metric and fluid terms. The third equation determines the radial fluid displacement W, driven by pressure perturbations X and metric variations. The fourth equation controls the pressure perturbation variable X, which is influenced by all other variables and encodes the fluid's response to spacetime curvature changes.

The remaining metric function H_0 and fluid function V are not independent but are determined algebraically by constraint equations that ensure consistency with

Einstein's field equations[?]:

$$
\begin{aligned}
H_0 = & \frac{1}{2b + \ell + Q} \Big\{ 8\pi r^2 e^{-\nu/2} X \\
& - \left[(\ell^2 + \ell - 1)Q - \omega^2 r^2 e^{-(\nu+\lambda)} \right] H_1 \\
& + \left[\ell(\ell+1) - \omega^2 r^2 e^{-\nu} - Q(Qe^{\lambda} - 1) \right] K \Big\}, \\
V = & \frac{e^{\nu/2}}{\omega^2} \left[\frac{X}{\epsilon + p} - \frac{Q}{r^2} e^{(\nu+\lambda)/2} W - e^{\nu/2} \frac{H_0}{2} \right].
\end{aligned}
\tag{2.47}
$$

These constraints reduce the number of dynamical degrees of freedom from four to two, consistent with the wave-like nature of the perturbations where only two polarizations of gravitational waves are possible.

The boundary conditions ensure physical regularity at the center ($r = 0$) and a free surface at the stellar boundary ($r = R$):

$$
\begin{aligned}
W(0) &= 1, \\
K(0) &= H_0(0), \\
H_1(0) &= \frac{\ell K(0) + 8\pi(\epsilon_0 + p_0) W(0)}{\ell(\ell+1)}, \\
X(0) &= (\epsilon_0 + p_0) e^{\nu_0/2} \left[\frac{4\pi}{3}(\epsilon_0 + 3p_0) - \frac{\omega^2}{\ell} e^{-\nu_0} \right] W(0) \\
&\quad + \frac{K(0)}{2},
\end{aligned}
\tag{2.48}
$$

with the surface condition $X(R) = 0$ ensuring vanishing Lagrangian pressure perturbation. This formulation captures the complete physics of nonradial oscillations in relativistic stars, providing the foundation for calculating quasinormal modes and their gravitational wave signatures.

2.9.1 The exterior region of the NS

In the vacuum region outside a neutron star, where matter perturbations vanish, the dynamics are governed solely by the degrees of freedom of the spacetime itself. The two independent metric perturbations, denoted by H_1 and K, can be combined into a single master variable satisfying a second-order wave equation known as the Zerilli equation [? ?]. This equation describes the propagation of gravitational waves in the exterior spacetime of the neutron star, defined as,

$$\left[\frac{d^2}{dr_*^2} + \omega^2 - V_Z(r)\right] Z(r) = 0, \tag{2.49}$$

where $Z(r)$ is the Zerilli function, ω is the complex frequency of the perturbation, and $V_Z(r)$ is the effective potentialgiven by,

$$\begin{aligned} V_Z(r) = &\frac{2\left(1 - \frac{2M}{r}\right)}{r^3(nr + 3M)^2} \\ &\times \left[n^2(n+1)r^3 + 3n^2Mr^2 + 9nM^2r + 9M^3\right], \end{aligned} \tag{2.50}$$

with $n = \frac{1}{2}(l-1)(l+2)$, where l is the angular quantum number, and M represents the total mass of the neutron star. The Zerilli potential $V_Z(r)$ is of fundamental importance as it encapsulates the entire influence of the curved spacetime on the propagating gravitational waves. Its shape creates a potential barrier surrounding the neutron star, which is peaked just outside the stellar radius. This barrier is responsible for several key physical phenomena: it partially traps gravitational radiation, leading to the existence of long-lived quasinormal modes; it filters and scatters incoming waves, determining the reflection and transmission coefficients for gravitational radiation; and its height and width directly influence the damping times and frequencies of the oscillations. The potential vanishes both at the stellar surface (approximated by the Schwarzschild radius in this exterior solution) and at spatial infinity, ensuring that wave solutions become simple plane waves in these asymptotic regions [**?**].

The tortoise coordinate r_* plays a crucial role in the analysis of wave propagation around neutron stars. It is defined by the transformation:

$$\frac{dr_*}{dr} = \frac{1}{1 - \frac{2M}{r}}, \tag{2.51}$$

which integrates to give the explicit form:

$$r_* = r + 2M \ln\left(\frac{r}{2M} - 1\right). \tag{2.52}$$

This coordinate transformation is particularly significant for neutron star physics as it regularizes the wave equation at the surface and beyond. While for black holes the tortoise coordinate maps the event horizon to negative infinity, for neutron stars (which lack an event horizon) it serves to simplify the wave equation and facilitate the implementation of boundary conditions [**?**]. The coordinate stretches the space around the compact object, providing a conformally flat background that is essential for cleanly separating incoming and outgoing wave solutions at infinity and ensuring proper treatment of the wave propagation in the strong-field region near the neutron star surface.

The solutions to the Zerilli equation represent gravitational wave modes in the neutron star's exterior. For general frequencies ω, the physical solution consists of a mixture of outgoing and ingoing waves at spatial infinity. The quasinormal modes correspond to specific discrete, complex frequencies ω_n that satisfy purely outgoing wave conditions at infinity. These modes characterize the natural oscillation frequencies of the neutron star spacetime, representing damped vibrations where gravitational wave emission carries energy away from the system. The real part of ω_n gives the oscillation frequency, while the imaginary part determines the damping rate due to gravitational wave emission [**? ?**].

2.9.2 f-mode eigenvalue problem

2.9.3 w-mode eigenvalue problem

Neutrinos interact with the ambient matter via weak interactions and it can modify the oscillation probabilities of neutrinos. The first experimental evidence of matter effects in neutrino oscillations came from the solar neutrino experiments, particularly the Sudbury Neutrino Observatory (SNO) [19] and Super-Kamiokande [20], around the early 2000s. These experiments confirmed the Mikheyev-Smirnov-Wolfenstein (MSW) effect [21–23], which describes how the oscillation probability of neutrinos gets modified as they travel through dense matter.

The modified Hamiltonian incorporating the neutrino-matter interaction can be written as,

$$\hat{H}_{eff} = \hat{H}_{vac} + \hat{H}_{matt}. \tag{2.53}$$

To have a close look at how the eigenvalues are modified under the influence of the neutrino-matter interaction, we take the help of the time-dependent Schrödinger equation, *i.e.*,

$$i\frac{d}{dt}\left|\nu(t)\right\rangle = \hat{H}_{eff}\left|\nu(t)\right\rangle . \tag{2.54}$$

This equation 2.54 can be rewritten in the flavor eigenstate as,

$$i\frac{d}{dt}\nu_\alpha(t) = \sum_\beta H_{\alpha\beta}\nu_\beta(t), \tag{2.55}$$

where α, β denote the flavor indices and $H_{\alpha\beta} = \langle\nu_\alpha|H|\nu_\beta\rangle$ are the matrix elements. On the other hand, $|\nu_k\rangle$ denotes the mass eigenstate and hence the eigenstate of the free Hamiltonian $\hat{H}_{vac}$, *i.e.*,

$$\hat{H}_{vac}\left|\nu_k\right\rangle = E_k\left|\nu_k\right\rangle , \tag{2.56}$$

where, E_k denotes the eigenvalue of the free Hamiltonian and can be written as $E_k = \sqrt{\vec{p}^2 + m_k^2}$, $\vec{p}$ being the momentum of the mass eigenstate $|\nu_k\rangle$. Neutrinos interact with the electrons of the ordinary matter through charged-current (CC) interaction via the $W^\pm$ boson of the SM. This interaction induces a matter potential confronting which the neutrino changes its flavor while propagating through a medium. This potential can be depicted as,

$$\begin{aligned} V_{CC,\alpha} &= \sqrt{2}G_F N_e, \quad [\alpha = e] \\ &= 0, \quad [\alpha = \mu, \tau]. \end{aligned} \tag{2.57}$$

Neutrinos mainly interact with the ambient electron of the matter while performing charged current interaction. There is another kind of interaction of neutrinos with the matter via the neutral mediator Z^0 boson of the SM, hence it is called neutral current (NC) interaction [24], the matter potential of which can be expressed as,

$$V_{NC,\alpha} = -\frac{G_F}{\sqrt{2}}N_n, \quad [\alpha = e, \mu, \tau], \tag{2.58}$$

,

where, N_e and N_n signify the electron number and neutron number densities of the matter, and G_F denotes the Fermi coupling constant ($G_F = 1.166 \times 10^{-5}$ GeV^{-2}).

Neutrinos interact with the matter via coherent and forward scattering [25]. Due to the scarcity of naturally produced muon and tau in the Earth matter, CC interaction only occurs with the electrons of the medium. Now, the CC interaction potential can be expressed in terms of some measurable parameters used in the experiments directly, *i.e.*, line-averaged Earth matter density ρ_{avg} [26] and the above-mentioned number densities (*i.e.*, N_e and N_n) as the following fashion,

$$V_{CC} \approx 7.6 \times \left(\frac{N_e}{N_p + N_n}\right) \times \left[\frac{\rho_{\text{avg}}}{10^{14}\,\text{g/cm}^3}\right] \text{eV}. \tag{2.59}$$

Considering further the medium of interaction is electrically neutral ($i.e., N_e = N_p$) and isoscaler ($i.e., N_p = N_n$), equation 2.59 can be rewritten as,

$$V_{CC} = 7.6 \times 0.5 \times \left[\frac{\rho_{\text{avg}}}{10^{14}\,\text{g/cm}^3}\right] \text{eV}. \tag{2.60}$$

The total matter potential $V(\nu)_\alpha = V_{CC,\alpha} + V_{NC,\alpha}$ changes its polarity while considering antineutrinos instead of neutrinos $i.e., V(\nu)_\alpha = -V(\bar{\nu})_\alpha$. This V_α is the eigenvalue of the Hamiltonian $\hat{H}_{\text{matt}}$ in weak basis, *i.e.*,

$$\hat{H}_{\text{matt}} \left|\nu_\alpha\right\rangle = V_\alpha \left|\nu_\alpha\right\rangle . \tag{2.61}$$

At tree level, neutral-current interactions mediated by the Z^0 boson are flavour universal, giving identical contributions to all active neutrinos, and hence it does not affect the oscillation probability. However, at the one-loop level, radiative corrections to the neutrino–Z^0 vertex involve virtual W bosons and the corresponding charged lepton partner. Since these corrections depend on the charged lepton mass, the large difference between m_μ and m_τ breaks flavor universality, leading to a small but finite splitting between the effective matter potentials of ν_μ and ν_τ [27]. The difference between the matter potential confronted by ν_μ and ν_τ can be written as [28],

$$\Delta V_{\tau\mu} = V_\tau - V_\mu = -\frac{3\sqrt{2}}{2\pi^2} G_F m_\tau^2 \left(ln\frac{m_W}{m_\tau} - \frac{1}{2}\right)\frac{N_e}{m_W^2}, \tag{2.62}$$

where, m_τ and m_W denote mass of tau and W boson. For Earth matter, the relative contribution of this splitting with respect to the potential $V_{\mathrm{CC,e}}$ is

$$\frac{\Delta V_{\tau\mu}}{V_{\mathrm{CC,e}}} \sim -2.5 \times 10^{-4}. \tag{2.63}$$

Although in standard interaction of active neutrinos, this effect is negligible, but from the context of dense environments like supernova, this slight difference between the interaction potentials becomes significant. It is the first non-zero splitting for the NC interaction of ν_μ and ν_τ (or in the 2-3 sector) which could affect neutrino oscillation in dense media.

From equation 2.56, it is evident that the energy of the neutrinos is the eigenvalues of the vacuum Hamiltonian in mass basis. To convert it into the physically measurable flavor basis, we need to diagonalize it through the unitary mixing matrix U by the following equation

$$\hat{H}^f_{vac} = U\hat{H}^m_{vac}U^\dagger. \tag{2.64}$$

. In the matrix form, it can be written as,

$$H^f = \mathbf{U}\begin{bmatrix} E^m_1 & 0 & 0 \\ 0 & E^m_2 & 0 \\ 0 & 0 & E^m_3 \end{bmatrix}\mathbf{U}^\dagger. \tag{2.65}$$

After invoking the influence of the matter effect in our picture, this mixing matrix U cannot diagonalize the effective Hamiltonian $\hat{H}_{eff}$, and it will not be diagonal even on the previous mass basis, considered so far. Hence, we need to construct a new type of mass basis (say, $|\nu^m_i\rangle$) where $\hat{H}_{vac}$ is diagonal. In this matter-modified mass basis $|\nu^m_i\rangle$, the diagonalized form of the effective Hamiltonian will be $\hat{H}^m = diag\,(E^m_1, E^m_2, E^m_3) = diag\,\frac{1}{2E}(\tilde{m}^2_1, \tilde{m}^2_2, \tilde{m}^3_3) = \tilde{U}\hat{H}_{eff}\tilde{U}^\dagger$. Here, $\tilde{U}$ is the matter-modified mixing matrix (kindly note that, $|\nu^m_i\rangle \neq |\nu_i\rangle\,, \tilde{U}^m \neq U$). Like the previous case, here also we assume equal momentum approximation (*i.e.*, $E_1 = E_2 = E_3 = E$) and $\tilde{m}^2_i$ signifies the matter-modified mass squared term. So, the flavor basis can now be expressed in terms of the matter-modified mass basis through this relation $|\nu_\alpha\rangle = \tilde{U}\,|\nu^m_i\rangle$. Adorned with this new mass basis and new mixing matrix, we can show the neutrino oscillation probability while neutrinos interact through matter with the two new plug-ins, $U_{\alpha i} \rightarrow \tilde{U}^m_{\alpha i}$ and $E_i \rightarrow E^m_i$, where E^m_i is the matter-modified energy eigenvalue of each mass state. Now, following the footprints of the same calculations mentioned in the section 2.5.1, we can write down the expression of the neutrino oscillation probability under the influence of the matter effect as,

$$P(\nu_\alpha \to \nu_\beta) = \delta_{\alpha\beta} - 4\sum_{i>j} Re(\tilde{U}^*_{\alpha i}\tilde{U}_{\beta i}\tilde{U}_{\alpha j}\tilde{U}^*_{\beta j}) \sin^2\left(\frac{\tilde{\Delta}_{ji}}{2}\right) + 2\sum_{i>j} Im(\tilde{U}^*_{\alpha i}\tilde{U}_{\beta i}\tilde{U}_{\alpha j}\tilde{U}^*_{\beta j}) \sin(\tilde{\Delta}_{ji}), \tag{2.66}$$

where, $\tilde{\Delta}_{ji} = \dfrac{\Delta\tilde{m}^2_{ji}L}{4E}$ denotes the modified frequency term with two measurable quantities, baseline length (L in km) and the neutrino energy (E in GeV).

CHAPTER 3

Dark matter admixed quarkyonic stars

3.0.1 Introduction

3.0.2 Results and Discussions

3.0.3 Conclusion

CHAPTER 4

f-mode Oscillation of Dark Matter Admixed Quarkyonic Star

4.0.1 Introduction

4.0.2 Results and Discussions

4.0.3 Conclusion

CHAPTER 5

w-mode Oscillation of Dark Matter Admixed Quarkyonic Star

5.0.1 Introduction

5.0.2 Results and Discussions

5.0.3 Conclusion

CHAPTER 6

Universality in Space-Time ω Modes of Quarkyonic Star

6.0.1 Introduction

6.0.2 Results and Discussions

6.0.3 Conclusion

CHAPTER 7

Summary and Conclusions

This thesis addresses some important issues in three-flavor neutrino oscillation in the context of upcoming high-precision long-baseline experiments. Neutrinos are omnipresent and they are the second most abundant particles in the Universe after photons. There are a variety of neutrino sources (both natural and artificial) producing neutrinos over a wide range of energies in the range of 10^{-4} to 10^{18} eV or so and traveling distances from a few meters to hundreds of Gpc. Neutrinos were postulated first by Wolfgang Pauli in 1930 to explain how beta-decay could conserve energy, momentum, and angular momentum. In 1934, Enrico Fermi named the new particle as "neutrino", which means the "little neutral one" in the Italian language. In 1956, Clyde L. Cowan and Frederick Reines first observed neutrinos in a nuclear reactor. There are three (3) light active flavor neutrinos: ν_e, ν_μ, and ν_τ - experimentally confirmed by the precision data on the Z-decay width at the e^+e^- collider at LEP [33]. They only take part in weak interactions through massive $W^\pm$ and Z^0 gauge bosons. They are electrically neutral, left-handed, and form $SU(2)$ doublets in the Standard Model with the corresponding left-handed charged leptons. In the basic SM of particle physics, neutrinos are massless due to the absence of right-handed neutrinos.

Over the past two decades, data from several world-class neutrino experiments firmly established that neutrinos change flavor after propagating a finite distance in space and time - a phenomenon known as "neutrino oscillation". Neutrino oscillation demands non-zero neutrino mass and mixing. Hence, non-zero neutrino mass is the first experimental proof for physics beyond the Standard Model for which the Nobel Prize was awarded to Takaaki Kajita and Arthur McDonald in 2015. Neutrino oscillation, in the 3ν paradigm, is mainly governed by six oscillation parameters, viz., solar mixing angle (θ_{12}), atmospheric mixing angle (θ_{23}), reactor mixing angle (θ_{13}), Dirac CP-violating phase ($\delta_{\rm CP}$), solar mass-squared splitting ($\Delta m^2_{21} \equiv m^2_2 - m^2_1$), and atmospheric mass-squared splitting ($\Delta m^2_{31} \equiv m^2_3 - m^2_1$). Still, there are a few fundamental issues that need to be resolved in neutrino oscillation experiments, namely, i) the neutrino mass ordering (whether $\Delta m^2_{31} > 0$, known as normal mass ordering - NMO, or $\Delta m^2_{31} < 0$, termed as inverted mass ordering - IMO), ii) the octant of θ_{23} (whether $\theta_{23} < 45^\circ$ known as lower octant - LO, or $\theta_{23} > 45^\circ$ termed as higher octant - HO), and iii) whether the CP is violated in the neutrino sector like quark sector and if so, then what is the precise value of the CP phase? The next-generation high-precision long-baseline (LBL) experiments are capable to address these issues at a high confidence level.

In this thesis, we study i) the physics reach of the upcoming LBL experiment, Deep Underground Neutrino Experiment (DUNE) in the US, to establish the possible deviation from maximal mixing of θ_{23}, ii) to achieve an improved precision on the 2-3 oscillation parameters using the synergy between the DUNE and Tokai-to-Hyper-Kamiokande (T2HK) experiments, and iii) the possible impact of the flavor-dependent long-range neutrino interactions on the measurements of neutrino oscillation parameters in DUNE and T2HK.

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Appendix

APPENDIX A

Phase Amplitude Method

A.0.1 Phase amplitude method

Equation (2.49) bears a formal resemblance to the time-independent Schrödinger equation; obtaining accurate solutions for quasinormal mode frequencies is a highly non-trivial numerical undertaking. The primary challenge lies in the precise implementation of the physical boundary condition requiring purely outgoing gravitational waves at spatial infinity. This requirement must be translated into a numerical computation in two problematic steps: first, one must approximate "infinity" by a finite but large radial coordinate, and second—constituting the major numerical difficulty—one must clearly separate two linearly independent solutions whose asymptotic behavior is exponentially growing and decaying, respectively. This numerical instability, where tiny errors in the decaying solution can be overwhelmed by contamination from the growing solution, has spawned the development of specialized techniques such as Leaver's continued-fraction method and various WKB/phase-integral approximations [**?** **?**].

In the present work, we employ the phase-amplitude method, as formulated by Anderson et al. [**?**]. This method is originally developed and demonstrated in the calculation of black hole quasinormal modes. Its application to the neutron star problem is well-motivated, as it directly addresses the core numerical difficulty. The method's advantage is highlighted in comparative studies where traditional phase-integral approaches, such as the one derived by Fröman et al.[**?**], were found to yield reliable frequencies only for the very lowest-order modes. In contrast, the phase-amplitude method is shown to generate highly accurate normal-mode frequencies across a wide spectrum. It achieves this robustness by reformulating the problem in terms of a phase function and its derivative, which remain well-behaved numerically even where the wavefunction itself is not. This stability allows for a more precise determination of the complex eigenfrequencies that satisfy the outgoing-wave boundary condition. The following section provides a detailed description of the phase-amplitude formalism and its specific implementation for calculating the quasinormal modes of neutron stars. The phase-amplitude method, as introduced by Anderson et al. [**?**], provides a robust framework for overcoming the numerical challenges inherent in solving the Zerilli equation for quasinormal modes. The method begins with a transformation of the dependent variable designed to simplify the asymptotic behavior of the solutions. Specifically, one defines a new function Ψ related to the Zerilli function Z by:

$$Z = \left(1 - \frac{2M}{r}\right)^{-1/2} \Psi. \tag{A.1}$$

This transformation removes the first derivative term from the resulting wave equation, yielding a Schrödinger-like form:

$$\left(\frac{d^2}{dr^2} + U(r)\right) \Psi = 0, \tag{A.2}$$

where the effective potential $U(r)$ incorporates the original Zerilli potential $V_Z(r)$ along with additional terms arising from the coordinate transformation. The key insight is to express the two linearly independent solutions $\Psi^{\pm}$ in a phase-amplitude form:

$$\Psi^{\pm} = q^{-1/2} \exp\left[\pm i \int^{r} q(\hat{r})\, d\hat{r}\right], \tag{A.3}$$

where the function $q(r)$ satisfies the nonlinear differential equation[**?**]:

$$\frac{1}{2q}\frac{d^2q}{dr^2} - \frac{3}{4q^2}\left(\frac{dq}{dr}\right)^2 + q^2 - U = 0. \tag{A.4}$$

Although this equation is nonlinear and appears more complex, it possesses significant numerical advantages. While the original wavefunction Ψ oscillates rapidly, especially for high frequencies, the function $q(r)$ is typically slowly varying. This slow variation makes $q(r)$ amenable to stable numerical integration. Initial conditions for q can be generated using the WKB approximation $q \approx \sqrt{U}$ at a large distance from the star, where U varies slowly.

The phenomenon of Stokes lines plays a crucial role in understanding the behavior of the solutions and must be accounted before numerical integration [**?**]. Stokes lines are curves in the complex plane emanating from turning points (zeros of U) where the asymptotic behavior of the WKB solutions changes discontinuously. When crossing a Stokes line, the coefficient of the subdominant exponential solution can change abruptly—a phenomenon known as the Stokes phenomenon [**?**]. This means that a linear combination of Ψ^+ and Ψ^- that represents the physical solution on one side of a Stokes line may not be valid on the other side.

To handle this properly, one must identify the appropriate anti-Stokes lines in the complex r-plane. Anti-Stokes lines are curves along which the phase integral $\int q\,dr$ is purely real, ensuring that the solutions remain oscillatory and bounded. For quasinormal modes with complex frequency ω, the optimal integration path is a straight line with slope given by $\tan\theta = -\mathrm{Im}\,\omega/\mathrm{Re}\,\omega$, which aligns with an anti-Stokes line. This path choice is essential for suppressing the exponential divergence of the outgoing wave solution and obtaining numerically stable results. The derivative along this path is computed using:

$$\frac{dq}{dr} = e^{-i\theta}\frac{dq}{d\rho}, \tag{A.5}$$

where ρ is the real distance along the integration path. The phase-amplitude method inherently accounts for the Stokes phenomenon by working with the slowly varying q-function, which remains smooth across Stokes lines, thereby avoiding the discontinuities that come from WKB approaches.

The numerical solution proceeds by integrating the q-equation from a large complex r (where the WKB initial conditions are valid) inward along the chosen anti-Stokes line to the stellar surface $r = R$. At the surface, the exterior solution must match the interior solution. The matching condition leads to an expression for the amplitude ratio of incoming to outgoing waves. This ratio can be expressed as [**?**]:

$$\mathcal{R}(\omega) = \frac{\mathcal{Z}_2 \left[\left(1 - \frac{2M}{R}\right) \left(iq + \frac{1}{2q}\frac{dq}{dr}\right) + \frac{M}{R^2} \right] + \frac{d\mathcal{Z}_2}{dr}}{\mathcal{Z}_2 \left[\left(1 - \frac{2M}{R}\right) \left(iq - \frac{1}{2q}\frac{dq}{dr}\right) - \frac{M}{R^2} \right] - \frac{d\mathcal{Z}_2}{dr}}, \tag{A.6}$$

where $\mathcal{Z}_2$ represents the interior solution evaluated at the stellar surface, and all quantities are computed at $r = R$. The quasinormal modes are precisely those complex frequencies ω_n for which this ratio vanishes, $\mathcal{R}(\omega_n) = 0$, indicating no incoming radiation. This condition is solved iteratively, and the use of the complex path along anti-Stokes lines ensures that the exponentially growing component is controlled, allowing for accurate determination of the mode frequencies. This approach has proven highly effective for both black hole and neutron star perturbations, providing reliable results even for highly damped modes where traditional WKB methods fail.

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